

+++ title = "" author = ["Emile Pierret"] draft = false +++

More details on selected publications from this list can be found in the *Featured Publications* section.

Preprints

- **Flow Matching for Applied Mathematicians.** Émile Pierret, *Valentine Tosel*, Julie Delon and Alasdair Newson, 2026. Submitted preprint. hal, `{{< bibtex id="PierretToselDelonNewsonFMforAM2026">}} @misc{PierretToselDelonNewsonFMforAM2026, author={Pierret, Emile and Tosel, Valentine and Delon, Julie and Newson, Alasdair}, booktitle={Submitted Preprint}, title={Flow Matching for Applied Mathematicians.}, year={2026} } {{< /bibtex >}}`
- **Exact Evaluation of the Accuracy of Diffusion Models for Inverse Problems with Gaussian Data Distributions.** Émile Pierret and Bruno Galerne, 2025. Submitted preprint. hal, arXiv, `{{< bibtex id="PierretGalerieEvaluationaccuracydiffusionmodels">}} @misc{PierretGalerieEvaluationaccuracydiffusionmodels, author={Pierret, 'Emile and Galerne, Bruno}, booktitle={Submitted Preprint}, title={Exact Evaluation of the Accuracy of Diffusion Models for Inverse Problems with {G}aussian Data Distributions}, year={2025} } {{< /bibtex >}}`

Conference publications

- **Tessellations of Semi-Discrete Flow Matching** Émile Pierret*, Johannes Hertrich*, Samuel Hurault* and Julie Delon, 2026. Accepted at Neurips 2026 as an oral. `{{< bibtex id="pierrethertrichhuraultdelontessellationssemidiscreteflowmatching2026">}} @inproceedings{pierrethertrichhuraultdelontessellationssemidiscreteflowmatching2026, title={Tessellations of Semi-Discrete Flow Matching}, author={Emile Pierret and Johannes Hertrich and Samuel Hurault and Julie Delon}, booktitle={The Fortieth Annual Conference on Neural Information Processing Systems}, year={2026} } {{< /bibtex >}}`
- **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.** Émile Pierret and Bruno Galerne, 2025. Accepted at ICML 2025. hal, arXiv, GitHub, `{{< bibtex id="PierretGalernediffusionmodelsGaussianexactsolutionerrorsICML2025">}} @inproceedings{PierretGalernediffusionmodelsGaussianexactsolutionerrorsICML2025, title={Diffusion models for {G}aussian distributions: Exact solutions and Wasserstein errors}, author={Emile Pierret and Bruno Galerne}, booktitle={Forty-second International Conference on Machine Learning}, year={2025}, } {{< /bibtex >}}`
- **Stochastic Super-Resolution for Gaussian textures.** Émile Pierret and Bruno Galerne, 2023. ICASSP 2023. hal, arXiv, IEEE preview, `{{< bibtex id="PierretGalernestochasticSRforgaussiantexturesICASSP2023">}} @inproceedings{PierretGalernestochasticSRforgaussiantexturesICASSP2023, author={Pierret, Émile and Galerne, Bruno}, booktitle={ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)}, title={Stochastic Super-Resolution For {G}aussian Textures}, year={2023}, volume={}, number={}, pages={1-5}, doi={10.1109/ICASSP49357}, } {{< /bibtex >}}`

Journal publication

- **Stochastic Super-Resolution for Gaussian microtextures.** Émile Pierret and Bruno Galerne, 2025. SIAM Journal on Imaging Sciences (SIIMS), Vol. 18, No. 2, pp. 1176–1207. hal, arXiv, GitHub, `{{< bibtex id="PierretGalernestochasticsuperresolutiongaussianmicrotexturesSIIMS2025">}}`

>}} @article{PierretGalernestochasticsuperresolutiongaussianmicrotexturesSIIMS2025, author = {Pierret, 'E}mile and Galerne, Bruno}, title = {Stochastic Super-resolution for {G}aussian Microtextures}, journal = {SIAM Journal on Imaging Sciences}, volume = {18}, number = {2}, pages = {1176-1207}, year = {2025}, doi = {10.1137/24M1657407}, } {{< /bibtex >}}